



JAD FALAQ

Generative AI Engineer

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6-month final-year engineering internship starting January 2027

PROFILE

Generative AI Engineer, final-year AI engineering student at **ENSIAS**, with experience at **Renault Digital** and **AKWA Group**. Specialized in **agentic AI**, **RAG**, **multi-agent systems** and **enterprise LLM integration**. Hybrid **generative AI / business** profile, focused on **QA process automation**, **sensitive data governance** and **production deployment**.

LLM – RAG – Agentic AI – MLOps – Full-stack Development

PROFESSIONAL EXPERIENCE

AI Engineer Intern – Quality Engineering

Renault Digital | Internship | Casablanca, Morocco

Jun. 2026 – Aug. 2026

- **AI Agents & Test Generation:** Designed and deployed to production a multi-agent LLM system (Requirement Analyzer Agent, Test Design Agent) turning requirements into Jira tickets and Gherkin scenarios, with human validation at each step (human-in-the-loop). POC on Copilot Studio and Power Platform, then full-stack rebuild (FastAPI, React/TypeScript). 100+ Jira tickets and 50 Gherkin scenarios generated, writing time cut from 10–20 min to 1–2 min per requirement, adopted by a 6-person QA team.
- **RAG & Document Ingestion:** Built a RAG on the Microsoft Copilot API (cosine similarity, top-5) fed by an ingestion pipeline of 400+ PDFs (Confluence, past user stories and Jira tickets): OCR of tables and images (unstructured.io), document-structure chunking and MMR retrieval on pgvector. DLP masking (Presidio, spaCy) before every LLM call.
- **Evaluation & Monitoring:** JSON generation validated by Pydantic schemas, then deterministic Gherkin rendering checked by the official Cucumber parser, eliminating LLM syntax errors. Similarity threshold calibration (0.72), token usage optimization and quality dashboard (85% of generations accepted without edits); 296 automated tests (pytest) and technical documentation of the 9 modules.
- **QA Automation:** Generation of Smoke/regression test files and Teams reporting (Power Automate); trained the engineer on the manual fallback.

ML & Data Engineer – Data Team

AKWA Group | Internship | Casablanca, Morocco

Jun. 2025 – Sep. 2025

- **Large-scale ML & Data:** Built an end-to-end ML pipeline (preprocessing, feature engineering, modeling, evaluation) detecting missing transactions across 6M+ SQL records, with automated ERP integration.
- **Model Evaluation:** Compared 5 algorithms (logistic regression, SVM, decision tree, Random Forest, KNN); the selected Random Forest reached 98% accuracy and 85% F1-score.

TECHNICAL SKILLS

- **LLM & Prompt Engineering:** system prompts, few-shot, JSON outputs, tool calling, OpenAI/Anthropic SDKs, Azure OpenAI, Microsoft Copilot API
- **Orchestration & Agents:** LangChain, LangGraph, ReAct agents, multi-agent systems, human-in-the-loop
- **RAG & Document Ingestion:** unstructured.io (OCR, table and image extraction), chunking (document-structure, recursive, semantic, agentic), embeddings, cosine similarity, pgvector, Chroma, vector and hybrid search, multi-query, RRF, MMR, reranking
- **Production & Back-end:** Python, JavaScript/TypeScript, C#, Node.js, FastAPI, Pydantic, REST APIs, SQLAlchemy, pytest, React, Firebase, Supabase
- **LLMOps & Evaluation:** LangSmith, Langfuse, RAG metrics, threshold calibration, quality monitoring, DLP (Presidio)
- **Machine Learning, Deep Learning & MLOps:** scikit-learn, PyTorch, YOLOv8, Computer Vision, Feature Engineering, Pandas, MLflow, DVC
- **Cloud & Deployment:** Docker, Git, GitHub Actions (CI/CD), Azure, Cloudflare

EDUCATION

State Engineering Degree – Artificial Intelligence Engineering (2IA)

ENSIAS, Rabat | Sep. 2024 – Jun. 2027

- Machine & Deep Learning (neural networks, reinforcement learning), statistical and probabilistic learning
- Big Data and distributed systems, data engineering, time series, multimedia processing (text, image)

Preparatory Classes for the French Grandes Écoles (CPGE)

CPGE Mohammed V, Casablanca | Sep. 2022 – Jun. 2024

- Applied mathematics, engineering sciences and Python programming

ACADEMIC PROJECTS

- **Generative AI & RAG:** RAG chatbot on Azure OpenAI (gpt-4.1-mini, text-embedding-3-small) comparing several retrieval strategies (vector, hybrid, multi-query, reranking), with Simple and Agentic RAG modes and performance metrics displayed
- **AI Agents & Orchestration:** LangGraph agent (LangChain, ReAct agents, AzureChatOpenAI) producing, from a job offer and a CV, a match analysis, a cover letter and interview questions, with application memory
- **NLP & Transformer Fine-tuning:** Fine-tuned AraBERT to detect medical misinformation in Arabic tweets (ArCOV19-Rumors), with deduplication, weighted loss, early stopping and threshold tuning, reaching 87% accuracy and 0.87 macro F1
- **Computer Vision & Deep Learning:** Detection of 7 dental pathologies (YOLOv8, 1,808 X-rays, 86% precision, MLflow, DVC)

SOFT SKILLS

- **Organization & Teamwork:** Logistics lead of the ENSIAS Forum Club (company–student meetings)
- **Autonomy & Problem Solving:** Delivered an AI solution to production under access and confidentiality constraints

LANGUAGES

French: Native – English: Bilingual – Spanish: B1